Sequence-level recursion in nonhuman vocal communication
A comparative test of five recursion types across 43 species

Jenny M. Michlich[1,2,3]

[1]Arizona State University, School of Complex Adaptive Systems, Tempe, USA
[2]Husson University, Department of Animal Behavior and Welfare, Bangor, USA
[3]Community College of Allegheny County, Department of Biology, Pittsburgh, USA

Abstract

A recurring question in comparative bioacoustics is whether nonhuman vocal sequences show recursive structure of the kind central to human language (Yip 2006). We tested five recursion types drawn from the phonological literature on recursive prosodic structure — center-embedding, tail-end embedding, left-boundary embedding, external (prefix) phrase expansion, and internal phrase expansion — against call-type sequences from 43 species of bats, birds, primates, a pinniped, and two cetaceans. Center-embedding, tail-end embedding, and left-boundary embedding were tested across 223 files; external and internal phrase expansion were tested across a subset of 82 files spanning nine species. Center-embedding was rare overall (5.8% of files); under Benjamini-Hochberg false discovery rate correction, two species retained significant positive results — yellow-naped amazon (Amazona auropalliata) duets and house wren (Troglodytes aedon) song — while bowhead and humpback whale showed raw-significant but FDR-non-surviving results, indicating weaker statistical support than an uncorrected reading would suggest. Gibbon song (three species, two genera, 30 sessions) showed no center-embedding, but Hylobates agilis showed significant external and internal phrase expansion in the majority of sessions, consistent with the progressive build-up structure documented for gibbon great calls. Bowhead and humpback whale, similar in their center-embedding results, diverged sharply on phrase expansion: humpback showed pervasive repetition across all sequence positions, bowhead did not. Tail-end embedding was the most broadly replicated pattern (49.8% of files, with 97 of 112 raw-significant results surviving FDR correction); a phylogenetically-corrected analysis found its association with vocal-learning category weakened but not eliminated relative to a naive estimate (PGLS slope = 1.01, $p = 0.121$, versus naive slope = 2.10, $p = 0.004$). The same phylogenetic treatment applied to external and internal phrase expansion was inconclusive, limited by the small number of species ($n = 6$) with a defined vocal-learning category. These findings are discussed in relation to the phonological recursion literature, in which recursive prosodic structure is documented as a real but non-default, minority-realized phenomenon even in human production.

Keywords: recursion; vocal learning; bioacoustics; phonology; sequence analysis; comparative cognition

Introduction

Recursion — the embedding of a structure within a structure of the same type — has occupied a central and contested place in discussions of what, if anything, is unique about human language (Hauser, Chomsky, and Fitch 2002). The question of whether phonological structure of this kind exists outside human language was posed directly by Yip (2006), whose call for a systematic search for phonology in other species motivates the comparative program pursued here. Center-embedded structure of the form A[B[C]B]A, in which an outer element brackets an inner one of the same category, is the paradigm case: it is well attested in human syntax and phonology, but claims of its presence in nonhuman communication remain sparse and contested. Rey, Perruchet, and Fagot (2012) demonstrated that Guinea baboons (Papio papio) could be trained to detect a center-embedded (AnBn) grammar in an operant task, raising the question of whether such structure emerges spontaneously in natural vocal sequences without training. A three-tier framework for vocal learning, distinguishing nonlearners, usage learners, and production learners (Vernes et al. 2021), provides an axis along which to ask whether sequence-level structural complexity scales with vocal-learning capacity. Recent work on self-embedded vocal motifs in wild orangutans (Lameira et al. 2023) and primate call rhythm (De Gregorio et al. 2025) has extended this comparative program to isochrony and rhythmic structure. Categorical treatment of call types, as adopted throughout this study, is independently supported in the songbird literature, where perception of call variants is categorical and context-dependent rather than continuous (Lachlan and Nowicki 2015).

The broader field of animal linguistics has recently been surveyed by Suzuki (2024), who identifies combinatorial and syntax-like structure in animal signals as an active area of interdisciplinary research. A landmark case is the chestnut-crowned babbler (Pomatostomus ruficeps), in which two acoustically distinct call elements are combined into different orderings associated with different behavioral contexts (Engesser et al. 2019). Bowling and Fitch (2015) caution against characterizing findings of this kind using linguistic terms such as phoneme, phonology, or syntax without qualification, since these terms are conventionally defined with reference to meaning, and meaning in animal communication is accessible only indirectly through behavioral inference. We follow this caution throughout: the analyses reported here concern statistically and acoustically defined call-type sequences, and terms such as recursion and phonological are used by structural analogy with the human linguistic literature that motivates the typology tested, not as a claim that the underlying units carry meaning in the linguistic sense. The quantitative, computational approach to analyzing animal vocal structure adopted here follows a tradition established by Clark, Marler, and Beeman (1987), who introduced computer-based methods for comparing and averaging animal vocalizations to quantify song structure and development in swamp sparrows (Melospiza georgiana).

This study addresses recursive structure within a single bout or continuous recording — the sequence of call-type units produced within one vocal sequence — referred to here as sequence-level or phonological-level recursion, by analogy with recursive structure among words or prosodic units within a single utterance in human language. Recursion at the level of the bout itself — whether whole bouts nest within larger bouts — is a distinct, structurally higher-order question not addressed here.

Five recursion types drawn from the phonological typology motivating this study were operationalized, illustrated in Figure 1. Center-embedding, tail-end embedding, and left-boundary embedding were tested at full scale. External and internal phrase expansion — concepts originating in Rothstein's (1989) and Koch's (1983) analysis of musical phrase structure and incorporated into the phonological recursion literature by Schreuder and Gilbers (2004; see also Schreuder n.d.) — were tested at a smaller scale on the subset of species most central to the findings reported here. Center-embedding is a symmetric bracketing pattern in which a sequence returns to its opening element after one or more internal elements (A…B…A). Tail-end embedding is a fixed unit repeated and iteratively appended to the end of a sequence (A[BC][BC][BC]…), the phonological analogue of what Schreuder terms tail-recursion, illustrated musically by repeated cadential motion. Left-boundary embedding is progressively deepening nested structure around a recurring anchor element, such that the distance between successive returns to the anchor grows over the course of a sequence. External (prefix) phrase expansion is the addition of subordinate, often repeated material before the main phrase, leaving the basic phrase itself unaffected; it was operationalized as excess repeated-bigram structure concentrated in the first third of a sequence relative to a permuted null. Internal phrase expansion is the addition of length within the basic phrase itself, typically via repetition; it was operationalized as excess repeated-bigram structure concentrated in the middle third of a sequence relative to a permuted null.

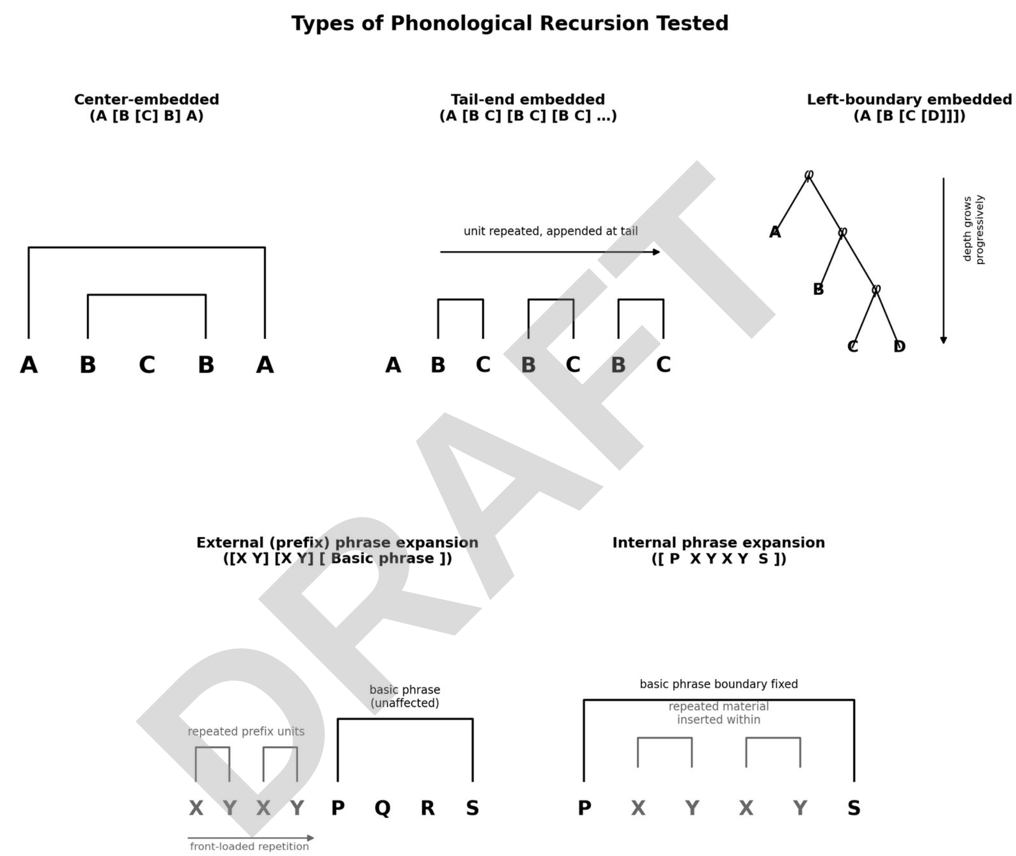

Figure 1. The five recursion types tested in this study: center-embedded, tail-end embedded, and left-boundary embedded (top row, tested at full scale, 223 files), and external (prefix) and internal phrase expansion (bottom row, tested on a subset of 82 files).

Materials and methods

Data sources

Call sequences were drawn from 43 species across five clades: Chiroptera (4 species), Aves (19 species), Primates (13 species, including one human vocal-percussion sample), Pinnipedia (1 species), and Cetacea (2 species), yielding 223 analyzable files for the three full-scale tests. Where available, call-type category labels were taken from expert-verified annotation, with data and companion publications cited jointly where both exist: Mediterranean monk seal vocal categories (Amlin et al. 2025; data provided as supplementary material within the published paper rather than a separate repository); bowhead whale song recordings (Stafford et al. 2018, Dryad dataset and companion paper); humpback whale phrase codes (Schall et al. 2021a, 2021b, Dryad dataset and companion paper), both from published Raven selection tables; yellow-naped amazon duet note-type codes (Dahlin et al. 2025, 2026, Dryad dataset and companion paper); house wren syllable measurements (Krieg 2024; Krieg and Wade 2023, Dryad dataset and companion paper); Hainan gibbon call-type codes (Dufourq et al. 2021, Zenodo dataset and companion paper); and Bengalese finch and wild white-rumped munia song recordings (Katahira et al. 2013, Zenodo dataset and companion paper). Hylobates agilis recordings were sourced from Morita and Koda (2019, Dryad dataset and companion paper); Nomascus gabriellae recordings from Clink et al. (2024, Zenodo dataset and companion bioRxiv preprint); Saccopteryx bilineata recordings from Fernandez et al. (2021, Dryad dataset and companion paper); Emballonura monticola and Emballonura alecto recordings from the ChiroVox bat call library (Görföl et al. 2022); Forpus passerinus recordings from Eggleston et al. (2022, Dryad dataset and companion paper); Papio papio recordings from Bonafos et al. (2023, Zenodo dataset) and the associated detection pipeline description (Bonafos et al. 2023); Phaethornis longirostris recordings from Araya-Salas and Wright (2013, Dryad dataset and companion paper); Rousettus aegyptiacus recordings from Prat et al. (2017, Figshare dataset and companion paper); Taeniopygia guttata recordings from Lee et al. (2026a, 2026b, Zenodo dataset and companion paper); and Human_beatbox recordings from the AVP-LVT vocal percussion dataset (Delgado Luezas 2021). The remaining 24 species were drawn from Xeno-canto (Xeno-canto Foundation), an online citizen-science database of wildlife sound recordings, using 172 recordings in total; call-type categories for these species were derived from raw audio via k-means clustering (k = 3) on four standardized acoustic features (duration, spectral centroid, spectral bandwidth, spectral rolloff) computed per detected call. Full per-recording attribution (recordist and catalog number) for all 172 Xeno-canto recordings is provided in Table 2. Onset and offset times were obtained from existing annotation or an energy-envelope detection pipeline, with all detections visually validated against spectrograms.

Statistical tests

For each file, the ordered sequence of call-type labels was tested against five statistics, each compared to a null distribution of 2000 random permutations of the same symbol sequence. Center-embedding was the count of immediate ABA triplets. Tail-end embedding was the count of immediately-repeated bigrams anywhere in the sequence. Left-boundary embedding was the trend in gap size between successive occurrences of the sequence's most frequent symbol. External and internal phrase expansion were each the count of immediately-repeated bigrams restricted to the first or middle third of the sequence respectively, compared to the same statistic on permuted sequences. All statistics were validated on synthetic random-order sequences to confirm an appropriately null result, and the two expansion statistics were additionally validated on synthetic sequences engineered with genuine front-loaded or middle-loaded repetition to confirm adequate statistical power. Given the large number of tests conducted, Benjamini-Hochberg false discovery rate (FDR) correction was applied separately within each of the three full-scale test types (center-embedding, tail-end embedding, boundary embedding) across all 223 files, at a corrected alpha of 0.05.

Phylogenetic comparative analysis

Species-level mean tail-end embedding z-scores were regressed on ordinal vocal-learning category (nonlearner = 0, usage learner = 1, production learner = 2) using ordinary least squares and phylogenetic generalized least squares (PGLS), the latter using a composite phylogeny built from published consensus divergence times spanning all sampled clades, to assess whether an apparent relationship between vocal-learning category and tail-end embedding reflects independent evolution rather than shared ancestry.

Results

Across 223 files, center-embedding was rare (5.8%), tail-end embedding was the most common pattern (50.2%), and left-boundary embedding was intermediate (12.1%; Table 1).

Table 1. Sequence-recursion test results by vocal-learning category (n=43 species, 223 files)

Vocal-learning category	Species (n)	Files (n)	Center-embedded significant	Tail-end embedded significant	Tail-end embedded mean z	Boundary embedded significant
Nonlearner	13	38	0/38 (0%)	10/38 (26%)	1.22	2/38 (5%)
Usage learner	11	71	4/71 (6%)	39/71 (55%)	2.94	11/71 (15%)
Production learner	13	76	4/76 (5%)	40/76 (53%)	3.50	10/76 (13%)
Production-learner candidate	3	25	5/25 (20%)	15/25 (60%)	6.50	2/25 (8%)
Unclassified	3	13	0/13 (0%)	8/13 (62%)	2.98	2/13 (15%)

Table 1. Sequence-recursion test results by vocal-learning category, 223 files across 43 species (center-embedding, tail-end embedding, and boundary embedding only).

Center-embedding

With 223 independent tests at $\alpha = 0.05$, approximately 11 significant results are expected by chance; 13 raw-significant results were observed. These concentrated in four species rather than scattering randomly: yellow-naped amazon parrot duets (3 of 6 duets), house wren song (3 of 18 songs), bowhead whale (2 of 14 recordings), and humpback whale (2 of 7 recordings). Three further species — Hapalemur occidentalis, wild Lonchura striata, and Rousettus aegyptiacus — showed exactly one significant file each, a pattern indistinguishable from the chance rate given the number of tests performed; these are treated as unresolved (Figure 2). Under Benjamini-Hochberg FDR correction, only 5 of the 223 tests remained significant, and of the four raw-replicated species, only parrot duets and house wren retained FDR-significant positive results; the bowhead and humpback whale results did not survive correction. This indicates that the parrot and house wren findings have stronger statistical support than the two whale findings, which should be treated as suggestive rather than confirmed.

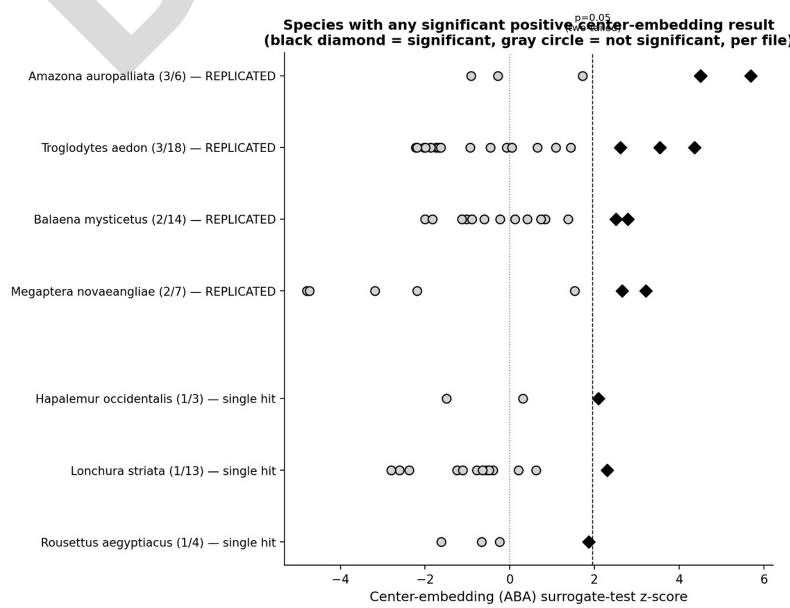

Figure 2. Species with any significant positive center-embedding result. Four species show multiple independent significant files; three show exactly one, consistent with chance.

Gibbon song

Three gibbon species spanning two genera were tested: Hylobates agilis (14 sessions, recorded at the Primate Research Institute, Kyoto University; Morita and Koda 2019), Nomascus gabriellae (5 sessions), and Nomascus hainanus (11 sessions, using call-type annotations from Dufourq et al. 2021). None of the 30 sessions showed significant center-embedding (Figure 3). This null result should not be read as agreement with the source literature for the H. agilis recordings: Morita and Koda (2019), analyzing the same recordings using a Bayesian grammar-induction approach, found non-regular (superregular) parses more probable than regular ones and concluded that the null hypothesis of shared grammatical structure between human language and gibbon song could not be rejected. The present center-embedding-specific null and Morita and Koda's broader superregularity finding are not in conflict: center-embedding is one specific, narrowly-defined pattern within the larger space of superregular structures, and its absence does not imply the absence of superregular structure more generally.

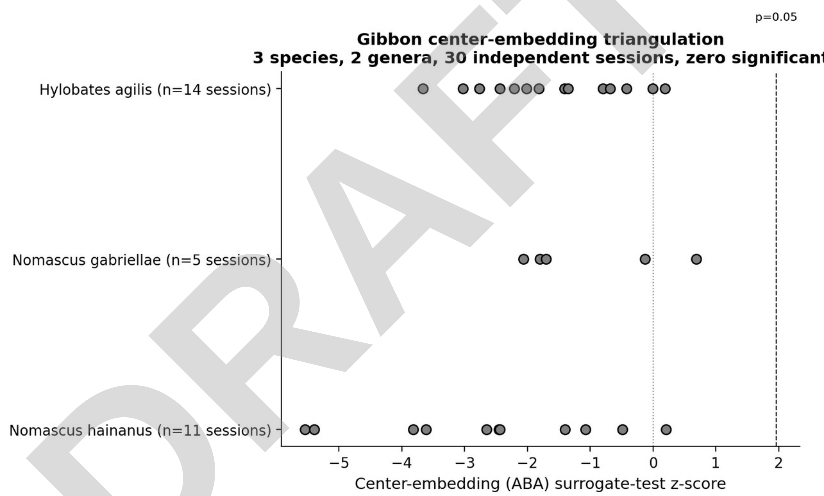

Figure 3. Center-embedding z-scores across 30 gibbon sessions spanning three species and two genera.

Hylobates agilis showed a different pattern for external and internal phrase expansion: 8 of 14 sessions were significant for external (prefix) expansion and 8 of 14 for internal expansion, with several large effect sizes (Figure 4). Nomascus hainanus showed a more modest signal (3 of 11 sessions for each test). Gibbon great-call song is characterized in the primary literature as a progressive build-up structure, with notes accelerating and intensifying toward a climax — an asymmetric, direction-dependent pattern consistent with external and internal expansion rather than the symmetric return that defines center-embedding. Gibbon song is accordingly better described as showing no center-embedding but substantial expansion-type structure, rather than as showing no recursive structure of any kind.

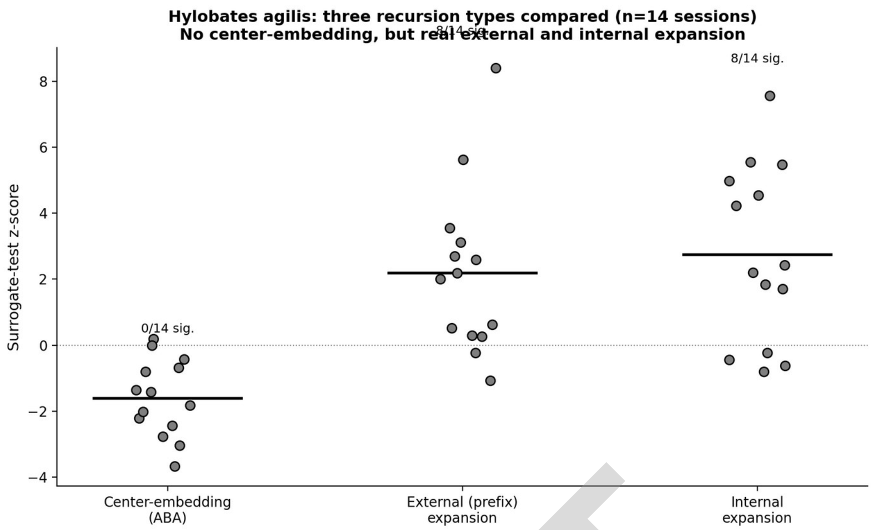

Figure 4. Hylobates agilis across all three tests: center-embedding is null; external and internal phrase expansion are both significant in the majority of sessions.

Species distinguishing center-embedding

The four raw-replicated species are each independently documented as having unusually rule-governed or hierarchically organized vocal sequences: parrot duets follow predetermined but flexible syntactic rules governing note order; house wren song is highly stereotyped and motif-repeating; baleen whale song is organized hierarchically into units, phrases, and themes. Of these, parrot and house wren additionally retain significance under FDR correction, while the two whale results should be treated more cautiously given they did not. Bengalese finch (Lonchura striata domestica), the standard example of complex finite-state song syntax in the ethological literature, showed the most uniformly negative center-embedding result of any species tested (17 files, mean z = −2.05, zero significant), though it showed modest external expansion (4 of 17 files). Finite-state sequential complexity and center-embedded recursion therefore appear to be distinct, separable properties of a vocal sequence (Figure 5).

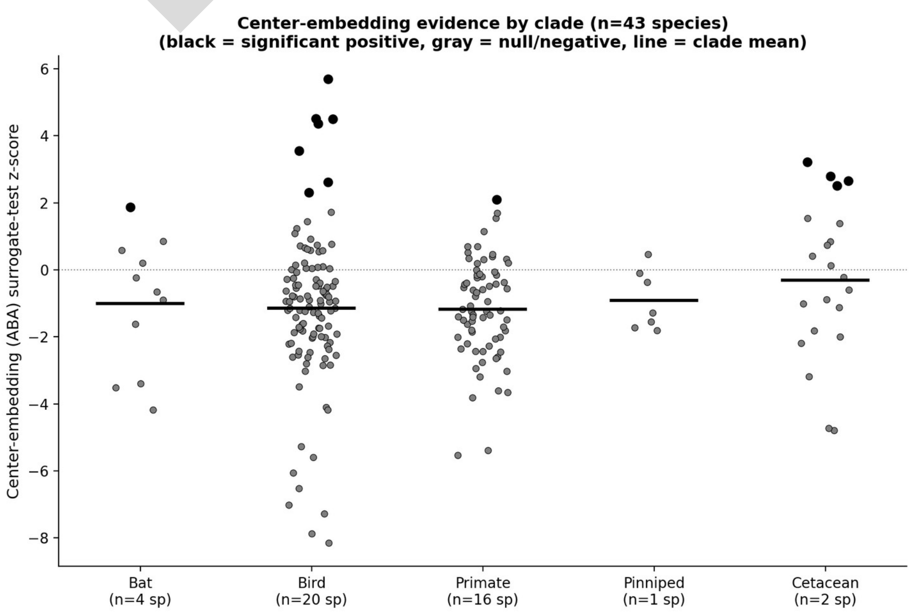

Figure 5. Center-embedding z-scores by clade, n = 43 species.

Tail-end embedding and vocal-learning category

Tail-end embedding was present in roughly half of all files in every clade and vocal-learning category (Figure 6), and the species-level distribution indicates this is a broad pattern across production-learner species rather than a small number of extreme outliers.

Figure 6. Tail-end embedding z-scores for all 43 species, grouped by vocal-learning category.

A naive regression treating species as statistically independent showed a significant positive relationship between vocal-learning category and tail-end embedding z-score (slope = 2.10, p = 0.004). Under phylogenetic correction this relationship weakened (PGLS slope = 1.01, p = 0.121), indicating that shared ancestry among closely related production-learner species accounts for part, but not all, of the naive association (Figure 7).

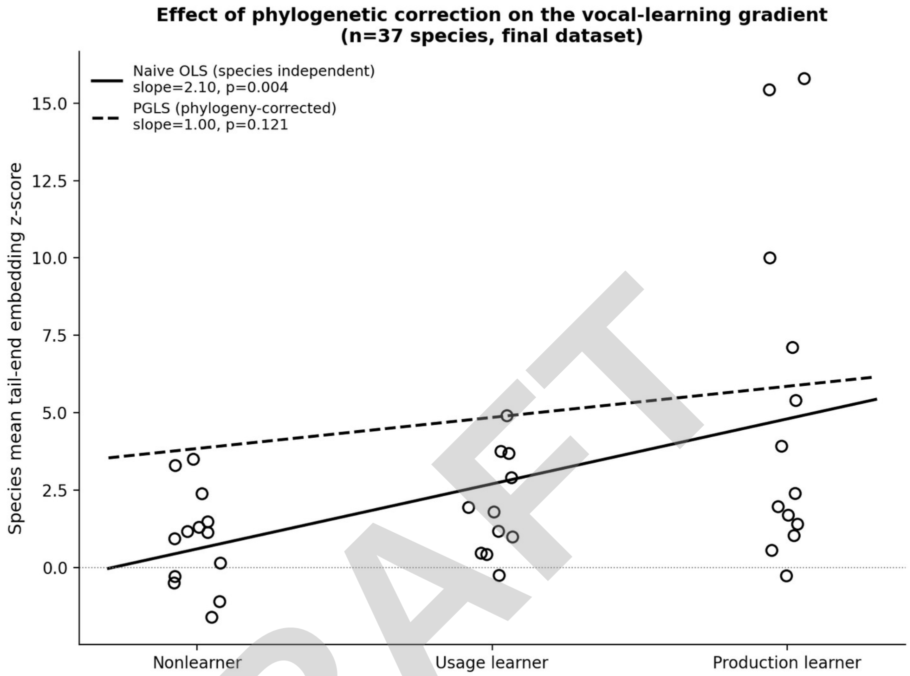

Figure 7. Naive OLS versus phylogenetically-corrected (PGLS) regression of tail-end embedding on vocal-learning category, n = 37 species.

External and internal phrase expansion

External and internal phrase expansion were tested across 82 files spanning the nine species central to the findings above. Both tests showed above-chance signal overall (35.4% and 25.6% of files significant respectively, against a 5% chance expectation; Figure 8).

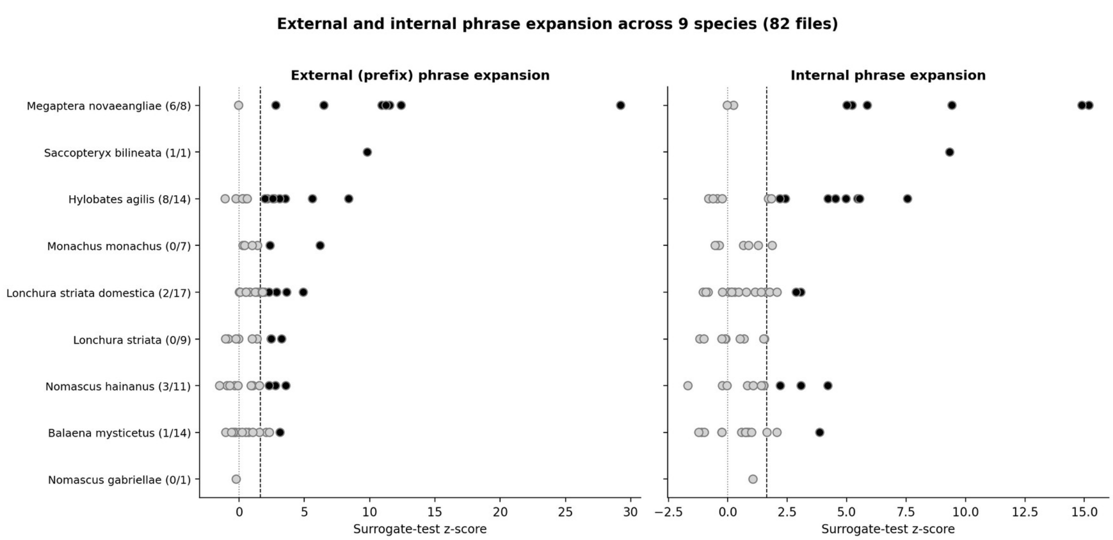

Figure 8. External (prefix) and internal phrase expansion across nine species, 82 files.

Bowhead and humpback whale, similar in their center-embedding results, diverged on phrase expansion. Humpback whale showed pervasive repetition across nearly every file and sequence position (7 of 8 files for external expansion, mean z = 10.6; 6 of 8 for internal expansion, mean z = 7.0; one file reaching z = 29.2). Bowhead whale showed almost no evidence of either expansion type (1 of 14 files for each, at the chance rate). Saccopteryx bilineata, tested on a single recording, showed extreme effects for both (z = 9.8 and 9.3). Baleen whale song does not therefore show a single, uniform recursive profile: bowhead and humpback appear to achieve center-embedding by different underlying means, and humpback shows a form of pervasive, sequence-wide repetition that bowhead does not share.

The same naive-versus-phylogenetically-corrected regression applied to tail-end embedding was repeated for external and internal expansion, restricted to the six species with a defined ordinal vocal-learning category. Neither test showed a significant relationship under either the naive or corrected model (external expansion: OLS p = 0.337, PGLS p = 0.364; internal expansion: OLS p = 0.594, PGLS p = 0.358). This analysis is underpowered at n = 6 species and should be treated as inconclusive rather than as evidence against a relationship.

Discussion

Schreuder, Gilbers, and Quéné's work on Dutch, the source of the boundary-embedding example in Figure 1 (the compound [onafhankelijk [Amsterdams [aardrijkskundig genoötschap]]]), reports that a recursive pitch-accent marker was realized on the embedded adjective in 22% of syntactically recursive phrases in production, against a 30% baseline rate of early accent even in non-recursive control phrases. The 22% figure is the appropriate benchmark for recursion-specific realization in that study. Our replicated species show center-embedding in a comparable minority range (parrot duets, 50%; house wren, 17%; bowhead whale, 14%; humpback whale, 29%). Elfner (2015) provides an independent empirical demonstration of recursive prosodic phrasing in production, in Connemara Irish. Together these sources indicate that phonological recursion, where directly measured in human production, is a real but minority, non-categorical phenomenon.

A related theoretical debate concerns the mechanism by which recursive prosodic phrasing arises. Constraint-based analyses distinguish ALIGN-type mappings, which align a single edge of a syntactic constituent with a prosodic phrase and rarely generate recursive structure in practice, from MATCH-type mappings, which align both edges symmetrically and do generate it (Féry 2023, following Selkirk 2011). Whether recursive phrasing is the analytically preferred default or a marked exception remains unresolved in this literature, paralleling the NoRecursion debate discussed below.

Cheng and Downing (2021) argue for a restrictive use of phonological recursion, showing that apparent recursive prosodic structure in several case-study languages is not syntactically motivated and can be reanalyzed without invoking recursion. They describe a formal NoRecursion constraint (Itô and Mester 2012; Selkirk 1995; Vigário 2010; Vogel 2009, 2012, 2019) under which recursive prosodic structure is a marked exception requiring independent motivation rather than a default assumption. The present finding that center-embedding is rare overall (5.8% of files) and concentrated in four of forty-three species, each independently linked to unusually rule-governed vocal behavior, is consistent with this view of recursion as an exceptional, independently-motivated property rather than a baseline expectation, a pattern that appears to hold within human prosodic phonology as well as across the taxa sampled here.

Absence of center-embedding does not indicate absence of recursive structure generally: Hylobates agilis shows no center-embedding but substantial external and internal phrase expansion, and the two baleen whale species tested show divergent expansion profiles despite similar center-embedding results. The five-type typology adopted here treats these as formally distinct constructs that need not co-occur, and the gibbon and whale results are consistent with that distinction.

The association between the four raw-replicated center-embedding species and independently-documented claims of syntactic or hierarchical organization was not designed in advance: no species was selected for testing on this basis, and all species were subject to an identical, pre-specified statistical procedure. This association is strongest for the two species whose results survive multiple-comparison correction, parrot duets and house wren; the whale results, while independently corroborated by descriptive literature on baleen whale song structure, currently rest on weaker statistical footing and warrant additional within-species replication before being treated as confirmed. The dissociation between the center-embedding pattern and general sequential complexity, illustrated by Bengalese finch, indicates that center-embedded recursion is separable from the finite-state syntactic complexity more commonly studied in birdsong.

The approach taken here, in which call-type sequences are subjected to statistical tests for structural pattern rather than manually inspected, is consistent with a broader recent shift toward data-driven discovery of phonetic and combinatorial structure in animal vocalization. Wang et al. (2024), for example, applied a self-supervised deep-learning model to discover phone-like units and grammar-like sequence patterns in domestic dog vocalizations without relying on hand-specified acoustic categories. The present analysis differs in method — acoustic clustering and permutation-based surrogate testing rather than deep learning — but shares the same underlying goal of letting structure in the data, rather than prior assumptions about what a call type should look like, determine the categories tested.

Acknowledgements

The author thanks colleagues who provided access to recordings and annotation data used in this analysis, and the many Xeno-canto recordists whose contributions made the Xeno-canto-sourced portion of this analysis possible (full attribution in Table 2).

Disclosure statement

No potential conflict of interest was reported by the author.

Data sources by species

Table 2 lists the data source and full citation for each of the 43 species analyzed. For the 24 species sourced from Xeno-canto, every individual recording's citation (recordist and catalog number) is listed.

Table 2. Data source and citation for each species.

Species	Clade	Source	Citation
Emballonura alecto	bat	ChiroVox	Görföl et al. 2022
Emballonura monticola	bat	ChiroVox	Görföl et al. 2022
Rousettus aegyptiacus	bat	Figshare	Prat et al. 2017
Saccopteryx bilineata	bat	Dryad	Fernandez et al. 2021
Amazona auropalliata	bird	Dryad	Dahlin et al. 2026
Carrion Crow	bird	Xeno-canto	Martin Billard, XC1011550. Accessible at www.xeno-canto.org/1011550. Eddy Scheinpflug, XC234879. Accessible at www.xeno-canto.org/234879. david m, XC341954. Accessible at www.xeno-canto.org/341954. Louis A. Hansen, XC458184. Accessible at www.xeno-canto.org/458184. Marie-Lan Taÿ Pamart, XC466390. Accessible at www.xeno-canto.org/466390. Seynaeve Adriaan, XC535993. Accessible at www.xeno-canto.org/535993. Stanislas Wroza, XC689520. Accessible at www.xeno-canto.org/689520. david m, XC844889. Accessible at www.xeno-canto.org/844889. Johannes Worona, XC977166. Accessible at www.xeno-canto.org/977166.
Columba livia	bird	Xeno-canto	Agris Celmins, XC1029257. Accessible at www.xeno-canto.org/1029257. Agris Celmins, XC1029258. Accessible at www.xeno-canto.org/1029258. Agris Celmins, XC1029262. Accessible at www.xeno-canto.org/1029262. Herman van der Meer, XC200010. Accessible at www.xeno-canto.org/200010. Timo Schnabel, XC270097. Accessible at www.xeno-canto.org/270097. Marie-Lan Taÿ Pamart, XC541143. Accessible at www.xeno-canto.org/541143. Peter Boesman, XC781620.

Species	Clade	Source	Citation
			Accessible at www.xeno-canto.org/781620.
Contopus virens	bird	Xeno-canto	Stanislas Wroza, XC1014933. Accessible at www.xeno-canto.org/1014933. Bobby Wilcox, XC1037302. Accessible at www.xeno-canto.org/1037302. Mario Reyes Jr, XC1135217. Accessible at www.xeno-canto.org/1135217. Greg Irving, XC429238. Accessible at www.xeno-canto.org/429238. Bates Estabrooks, XC429616. Accessible at www.xeno-canto.org/429616. Jerome Fischer, XC544558. Accessible at www.xeno-canto.org/544558. Russ Wigh, XC563444. Accessible at www.xeno-canto.org/563444. Russ Wigh, XC563447. Accessible at www.xeno-canto.org/563447. Christopher McPherson, XC691330. Accessible at www.xeno-canto.org/691330.
Coturnix japonica	bird	Xeno-canto	Alex Thomas, XC329204. Accessible at www.xeno-canto.org/329204. Stanislas Wroza, XC485730. Accessible at www.xeno-canto.org/485730. James Lidster, XC527210. Accessible at www.xeno-canto.org/527210. Christoph Bock, XC61928. Accessible at www.xeno-canto.org/61928. Christoph Bock, XC61942. Accessible at www.xeno-canto.org/61942. Oriental Stork, XC790229. Accessible at www.xeno-canto.org/790229. Oriental Stork, XC790231. Accessible at www.xeno-canto.org/790231. Goro, XC913389. Accessible at www.xeno-canto.org/913389.

Species	Clade	Source	Citation
Forpus passerinus	bird	Dryad	Eggleston et al. 2022
Gallus gallus	bird	Xeno-canto	Lim Ying Hien, XC1009091. Accessible at www.xeno-canto.org/1009091. Ding Li Yong, XC1067414. Accessible at www.xeno-canto.org/1067414. Aku Kalliomäki, XC1153675. Accessible at www.xeno-canto.org/1153675. Aku Kalliomäki, XC1153683. Accessible at www.xeno-canto.org/1153683. Marc Anderson, XC314766. Accessible at www.xeno-canto.org/314766. Marc Anderson, XC314767. Accessible at www.xeno-canto.org/314767. Peter Boesman, XC472720. Accessible at www.xeno-canto.org/472720. Olivier SWIFT, XC577805. Accessible at www.xeno-canto.org/577805. Olivier SWIFT, XC577813. Accessible at www.xeno-canto.org/577813. Olivier SWIFT, XC577814. Accessible at www.xeno-canto.org/577814. Ding Li Yong, XC689265. Accessible at www.xeno-canto.org/689265.
Lonchura striata	bird	Zenodo	Katahira et al. 2013
Lonchura striata domestica	bird	Zenodo	Katahira et al. 2013
Melopsittacus undulatus	bird	Xeno-canto	Peter Boesman, XC1066690. Accessible at www.xeno-canto.org/1066690. Peter Boesman, XC1066691. Accessible at www.xeno-canto.org/1066691. Marc Anderson, XC134241. Accessible at www.xeno-canto.org/134241. Marc Anderson, XC172266. Accessible at www.xeno-canto.org/172266. Mark Harper, XC40579. Accessible at www.xeno-canto.org/40579. Marc Anderson, XC593345.

Species	Clade	Source	Citation
			Accessible at www.xeno-canto.org/593345. Marc Anderson, XC609245. Accessible at www.xeno-canto.org/609245. Marc Anderson, XC611286. Accessible at www.xeno-canto.org/611286. nick talbot, XC820503. Accessible at www.xeno-canto.org/820503. Peter Boesman, XC859963. Accessible at www.xeno-canto.org/859963. Louis Masarei, XC924266. Accessible at www.xeno-canto.org/924266. Samantha Aguirre, XC940888. Accessible at www.xeno-canto.org/940888.
Menura novaehollandiae	bird	Xeno-canto	Andrew McCafferty, XC1040676. Accessible at www.xeno-canto.org/1040676. Marc Anderson, XC1045882. Accessible at www.xeno-canto.org/1045882. Romuald Mikusek, XC1103947. Accessible at www.xeno-canto.org/1103947. Romuald Mikusek, XC1163855. Accessible at www.xeno-canto.org/1163855. Romuald Mikusek, XC1163856. Accessible at www.xeno-canto.org/1163856. Romuald Mikusek, XC1172718. Accessible at www.xeno-canto.org/1172718. Liam Manderson, XC605887. Accessible at www.xeno-canto.org/605887. Zebedee Muller, XC804368. Accessible at www.xeno-canto.org/804368.
Mimus polyglottos	bird	Xeno-canto	noah eckman, XC1071700. Accessible at www.xeno-canto.org/1071700. Scott Crabtree, XC1089514.

Species	Clade	Source	Citation
			Accessible at www.xeno-canto.org/1089514. Ron Overholtz, XC559555. Accessible at www.xeno-canto.org/559555. Manuel Grosselet, XC798489. Accessible at www.xeno-canto.org/798489. Paul Marvin, XC798536. Accessible at www.xeno-canto.org/798536. Scott Gravette, XC871912. Accessible at www.xeno-canto.org/871912. Scott Olmstead, XC872514. Accessible at www.xeno-canto.org/872514. Étienne Leroy, XC958237. Accessible at www.xeno-canto.org/958237. Étienne Leroy, XC958238. Accessible at www.xeno-canto.org/958238. FREDERIC Lionel, XC997254. Accessible at www.xeno-canto.org/997254. FREDERIC Lionel, XC997262. Accessible at www.xeno-canto.org/997262. FREDERIC Lionel, XC997263. Accessible at www.xeno-canto.org/997263.
Phaethornis longirostris	bird	Dryad	Araya-Salas & Wright 2013
Psittacus erithacus	bird	Xeno-canto	Matthew L. Brady, XC1011340. Accessible at www.xeno-canto.org/1011340. Matthew L. Brady, XC1011809. Accessible at www.xeno-canto.org/1011809. Matthew L. Brady, XC1011810. Accessible at www.xeno-canto.org/1011810. Bernard BOUSQUET, XC1053186. Accessible at www.xeno-canto.org/1053186. Jonathan Onongo, XC1105395. Accessible at www.xeno-canto.org/1105395.

Species	Clade	Source	Citation
			Jonathan Onongo, XC1105396. Accessible at www.xeno-canto.org/1105396. Bram Vogels, XC404711. Accessible at www.xeno-canto.org/404711. A. Bennett Hennessey, XC599239. Accessible at www.xeno-canto.org/599239. Étienne Leroy, XC877895. Accessible at www.xeno-canto.org/877895.
Screaming Piha	bird	Xeno-canto	Frank Lambert, XC1043152. Accessible at www.xeno-canto.org/1043152. Herman van Oosten, XC152497. Accessible at www.xeno-canto.org/152497. Peter Boesman, XC225167. Accessible at www.xeno-canto.org/225167. Peter Boesman, XC225170. Accessible at www.xeno-canto.org/225170. Jeremy Minns, XC241664. Accessible at www.xeno-canto.org/241664. GABRIEL LEITE, XC303181. Accessible at www.xeno-canto.org/303181. JAYRSON ARAUJO DE OLIVEIRA, XC504274. Accessible at www.xeno-canto.org/504274. Jarrod Swackhamer, XC521303. Accessible at www.xeno-canto.org/521303. Dante Buzzetti, XC668412. Accessible at www.xeno-canto.org/668412. JAYRSON ARAUJO DE OLIVEIRA, XC978265. Accessible at www.xeno-canto.org/978265.
Serinus canaria	bird	Xeno-canto	Alain Malengreau, XC1056183. Accessible at www.xeno-canto.org/1056183. Lionel FREDERIC, XC1145741. Accessible at www.xeno-canto.org/1145741.

Species	Clade	Source	Citation
			Cedric Mroczko, XC262763. Accessible at www.xeno-canto.org/262763.
			Cedric Mroczko, XC316549. Accessible at www.xeno-canto.org/316549.
			Jacob Lotz, XC398574. Accessible at www.xeno-canto.org/398574.
			Baltasar, XC487915. Accessible at www.xeno-canto.org/487915.
			Rubén Barone, XC601672. Accessible at www.xeno-canto.org/601672.
			Rubén Barone, XC782280. Accessible at www.xeno-canto.org/782280.
			Carlos Pereira, XC907123. Accessible at www.xeno-canto.org/907123.
Taeniopygia guttata	bird	Zenodo	Lee et al. 2026a
			Lee et al. 2026b
Troglodytes aedon	bird	Dryad	Krieg & Wade 2023
			Krieg 2024
Turdus merula	bird	Xeno-canto	Ruslan Mazuryk, XC306263. Accessible at www.xeno-canto.org/306263.
			david m, XC310974. Accessible at www.xeno-canto.org/310974.
			Louis A. Hansen, XC382840. Accessible at www.xeno-canto.org/382840.
			Kieron Palmer, XC555637. Accessible at www.xeno-canto.org/555637.
			Cedric Mroczko, XC731207. Accessible at www.xeno-canto.org/731207.
			Albert Lastukhin, XC735931. Accessible at www.xeno-canto.org/735931.
			Esperanza Poveda, XC825630. Accessible at www.xeno-canto.org/825630.
			João Tomás, XC862600. Accessible at www.xeno-canto.org/862600.
			Johannes Worona, XC926144. Accessible at www.xeno-canto.org/926144.

Species	Clade	Source	Citation
			Catalin Ionescu, XC960501. Accessible at www.xeno-canto.org/960501.
			Catalin Ionescu, XC985000. Accessible at www.xeno-canto.org/985000.
			Albert Dees, XC995330. Accessible at www.xeno-canto.org/995330.
Zenaida macroura	bird	Xeno-canto	Stanislas Wroza, XC1017870. Accessible at www.xeno-canto.org/1017870.
			Paul Marvin, XC153653. Accessible at www.xeno-canto.org/153653.
			Richard E. Webster, XC583578. Accessible at www.xeno-canto.org/583578.
			Peter Ward and Ken Hall, XC613539. Accessible at www.xeno-canto.org/613539.
			Thomas Ryder Payne, XC636552. Accessible at www.xeno-canto.org/636552.
			Christopher McPherson, XC638304. Accessible at www.xeno-canto.org/638304.
			Alán Palacios, XC730163. Accessible at www.xeno-canto.org/730163.
			Richard E. Webster, XC940517. Accessible at www.xeno-canto.org/940517.
			Richard E. Webster, XC940522. Accessible at www.xeno-canto.org/940522.
Balaena mysticetus	cetacean	Dryad	Stafford et al. 2018
Megaptera novaeangliae	cetacean	Dryad	Schall et al. 2021a
			Schall et al. 2021b
Monachus monachus	pinniped	paper supplement	Amlin et al. 2025
Callithrix aurita	primate	Xeno-canto	Marcelo J Feliti, XC1033128. Accessible at www.xeno-canto.org/1033128.
			Guilherme Melo, XC1166629. Accessible at www.xeno-canto.org/1166629.
			Guilherme Melo, XC1166630. Accessible at www.xeno-canto.org/1166630.
			Guilherme Melo,

Species	Clade	Source	Citation
			XC1166631. Accessible at www.xeno-canto.org/1166631.
Cebus malitiosus	primate	Xeno-canto	Guilherme Melo, XC1106858. Accessible at www.xeno-canto.org/1106858. Guilherme Melo, XC1173134. Accessible at www.xeno-canto.org/1173134.
Cercopithecus diana	primate	Xeno-canto	Étienne Leroy, XC960201. Accessible at www.xeno-canto.org/960201. Étienne Leroy, XC960202. Accessible at www.xeno-canto.org/960202.
Chlorocebus pygerythrus	primate	Xeno-canto	Marc Anderson, XC1039879. Accessible at www.xeno-canto.org/1039879. Alain Malengreau, XC1089539. Accessible at www.xeno-canto.org/1089539. Stein Ø. Nilsen, XC961506. Accessible at www.xeno-canto.org/961506. Stein Ø. Nilsen, XC961507. Accessible at www.xeno-canto.org/961507.
Eulemur rubriventer	primate	Xeno-canto	Frank Lambert, XC1072253. Accessible at www.xeno-canto.org/1072253. Frank Lambert, XC1072254. Accessible at www.xeno-canto.org/1072254. Frank Lambert, XC1072255. Accessible at www.xeno-canto.org/1072255. Albert Lastukhin, XC983966. Accessible at www.xeno-canto.org/983966. Albert Lastukhin, XC983967. Accessible at www.xeno-canto.org/983967.
Hapalemur occidentalis	primate	Xeno-canto	Frank Lambert, XC1081907. Accessible at www.xeno-canto.org/1081907. Frank Lambert, XC1081908. Accessible at www.xeno-canto.org/1081908. Frank Lambert, XC1081909. Accessible at www.xeno-

Species	Clade	Source	Citation
			canto.org/1081909.
Human beatbox	primate	Zenodo	Delgado et al. 2019 Delgado Luezas 2021
Hylobates agilis	primate	Dryad	Morita & Koda 2019
Macaca cyclopis	primate	Xeno-canto	Lars Lachmann, XC1045268. Accessible at www.xeno-canto.org/1045268. Lars Lachmann, XC1045270. Accessible at www.xeno-canto.org/1045270. Lars Lachmann, XC1045271. Accessible at www.xeno-canto.org/1045271. Sheng-wun Jheng, XC1139469. Accessible at www.xeno-canto.org/1139469. Sheng-wun Jheng, XC985977. Accessible at www.xeno-canto.org/985977.
Macaca fascicularis	primate	Xeno-canto	Frank Lambert, XC1031578. Accessible at www.xeno-canto.org/1031578. Daan Drukker, XC1063259. Accessible at www.xeno-canto.org/1063259. Simon Elliott, XC960848. Accessible at www.xeno-canto.org/960848.
Macaca mulatta	primate	Xeno-canto	Swami Bogim, XC1006942. Accessible at www.xeno-canto.org/1006942. Ding Li Yong, XC1044523. Accessible at www.xeno-canto.org/1044523. Dave Guo, XC1047250. Accessible at www.xeno-canto.org/1047250. Marc Anderson, XC1072576. Accessible at www.xeno-canto.org/1072576. Marc Anderson, XC960761. Accessible at www.xeno-canto.org/960761.
Nomascus gabriellae	primate	Zenodo	Clink et al. 2024
Nomascus hainanus	primate	Zenodo	Dufourq et al. 2021
Pan troglodytes	primate	Xeno-canto	Antoine Ede, XC1036135. Accessible at www.xeno-canto.org/1036135. Budongo Conservation Field Station, XC1054154.

Species	Clade	Source	Citation
			Accessible at www.xeno-canto.org/1054154. Budongo Conservation Field Station, XC1054178. Accessible at www.xeno-canto.org/1054178. Budongo Conservation Field Station, XC1070341. Accessible at www.xeno-canto.org/1070341. Budongo Conservation Field Station, XC1070343. Accessible at www.xeno-canto.org/1070343. Budongo Conservation Field Station, XC1070346. Accessible at www.xeno-canto.org/1070346. Budongo Conservation Field Station, XC1070347. Accessible at www.xeno-canto.org/1070347. Jonathan Onongo, XC1140926. Accessible at www.xeno-canto.org/1140926. Étienne Leroy, XC960121. Accessible at www.xeno-canto.org/960121. Étienne Leroy, XC960498. Accessible at www.xeno-canto.org/960498.
Papio papio	primate	Zenodo	Bonafos et al. 2023
Pongo pygmaeus	primate	Xeno-canto	Marc Anderson, XC1097762. Accessible at www.xeno-canto.org/1097762. Daan Drukker, XC959719. Accessible at www.xeno-canto.org/959719. Marc Anderson, XC960775. Accessible at www.xeno-canto.org/960775. GABRIEL LEITE, XC985671. Accessible at www.xeno-canto.org/985671.

References

Amlin, A., L. Rendell, N. Pierantonio, C. Andrés-Hervías, G. Notarbartolo di Sciara, G. Hastie, and J. Gonzalvo. 2025. “Confirmed Vocal Activity of Mediterranean Monk Seals in the Inner Ionian Sea Archipelago, Greece.” Journal of the Acoustical Society of America 157 (4): 3204–3213.

Araya-Salas, M., and T. Wright. 2013. “Data from: Open-Ended Song Learning in a Hummingbird.” Dataset. Dryad.

Araya-Salas, M., and T. Wright. 2013. “Open-Ended Song Learning in a Hummingbird.” Biology Letters 9 (5): 20130625.

Berg, K., and R. Eggleston. 2022. “Vocal Babbling in a Wild Parrot Shows Life History and Endocrine Affinities with Human Infants.” Dataset. Dryad.

Bonafos, G., P. Pudlo, J. M. Freyermuth, T. Legou, J. Fagot, S. Tronçon, and A. Rey. 2023. “Detection and Classification of Vocal Productions in Large Scale Audio Recordings.” arXiv:2302.07640.

Bonafos, G., P. Pudlo, J. M. Freyermuth, T. Legou, J. Fagot, S. Tronçon, and A. Rey. 2023. “Guinea Baboon Vocalizations Dataset Automatically Extracted with a Deep Neural Network from Natural Audio Recordings.” Dataset. Zenodo.

Bowling, D. L., and W. T. Fitch. 2015. “Do Animal Communication Systems Have Phonemes?” Trends in Cognitive Sciences 19 (10): 555–557.

Cheng, L. L.-S., and L. J. Downing. 2021. “Recursion and the Definition of Universal Prosodic Categories.” Languages 6 (1): 25.

Clark, C. W., P. Marler, and K. Beeman. 1987. “Quantitative Analysis of Animal Vocal Phonology: An Application to Swamp Sparrow Song.” Ethology 76: 101–115.

Clink, D. J., H. Cross-Jaya, J. Kim, A. H. Ahmad, M. Hong, R. Sala, H. Birot, C. Agger, T. T. Vu, H. N. Thi, and T. N. Chi. 2024. “Benchmarking for the Automated Detection and Classification of Southern Yellow-Cheeked Crested Gibbon Calls from Passive Acoustic Monitoring Data.” bioRxiv 2024:2024-08.

Clink, D., H. Cross-Jaya, J. Kim, A. H. Ahmad, M. Hong, R. Sala, H. Birot, C. Agger, T. T. Vu, H. N. Thi, T. N. Chi, and H. Klinck. 2024. “Dataset for ‘Benchmarking for the Automated Detection of Southern Yellow-Cheeked Crested Gibbon Calls from Passive Acoustic Monitoring Data.’” Dataset. Zenodo.

Dahlin, C. R., B. Killinger, M. Snipe, O. Small, B. Guenther, and L. Richwine. 2025. “Data from: Decoding Parrot Duets: Complex Communication in Yellow-Naped Amazons.” Dataset. Dryad.

Dahlin, C. R., B. Killinger, M. Snipe, O. Small, B. Guenther, and L. Richwine. 2026. “Decoding Parrot Duets: Complex Communication in Yellow-Naped Amazons.” Journal of Avian Biology 2026: e03552.

De Gregorio, C., et al. 2025. “Isochrony and Rhythmic Call Structure Across Primate Taxa.” Annals of the New York Academy of Sciences.

Delgado Luezas, A. 2021. “AVP-LVT Vocal Percussion Dataset.” Dataset. Zenodo.

Delgado, A., S. McDonald, N. Xu, and M. Sandler. 2019. “A New Dataset for Amateur Vocal Percussion Analysis.” In Proceedings of the 14th International Audio Mostly Conference: A Journey in Sound (AM ’19), 17–23. New York, NY: Association for Computing Machinery.

Dufourq, E., et al. 2021. “Data for: Automated Detection of Hainan Gibbon Calls for Passive Acoustic Monitoring.” Dataset. Zenodo.

Dufourq, E., I. Durbach, J. P. Hansford, A. Hoepfner, H. Ma, J. V. Bryant, C. S. Stender, W. Li, Z. Liu, Q. Chen, Z. Zhou, and S. T. Turvey. 2021. “Automated Detection of Hainan Gibbon Calls for Passive Acoustic Monitoring.” Remote Sensing in Ecology and Conservation.

Eggleston, R., N. Viloria, S. Delgado, A. Mata, H. Y. Guerrero, R. J. Kline, S. R. Beissinger, and K. S. Berg. 2022. “Vocal Babbling in a Wild Parrot Shows Life History and Endocrine Affinities with Human Infants.” Proceedings of the Royal Society B 289 (1976): 20220592.

Elfner, E. 2015. “Recursion in Prosodic Phrasing: Evidence from Connemara Irish.” Natural Language and Linguistic Theory.

Engesser, S., et al. 2019. “Combinatoriality and Compositionality in the Vocal System of Chestnut-Crowned Babblers.” Proceedings of the National Academy of Sciences 116 (14): 6046–6051.

Féry, C. 2023. “Intermediate Phonology, Part 7: Prosodic Phrase and Intonation Phrase.” CreteLing course materials.

Fernandez, A. A., L. S. Burchardt, M. Nagy, and M. Knörnschild. 2021. “Babbling in a Vocal Learning Bat Resembles Human Infant Babbling.” Science 373 (6557): 923–926.

Fernandez, A. A., L. S. Burchardt, M. Nagy, and M. Knörnschild. 2021. “Data from: Babbling in a Vocal Learning Bat Resembles Human Infant Babbling.” Dataset. Dryad.

Görföl, T., J. C.-C. Huang, G. Csorba, D. Győrössy, P. Estók, T. Kingston, K. L. Szabadi, et al. 2022. “ChiroVox: A Public Library of Bat Calls.” PeerJ 10: e12445.

Hauser, M. D., N. Chomsky, and W. T. Fitch. 2002. “The Faculty of Language: What Is It, Who Has It, and How Did It Evolve?” Science 298 (5598): 1569–1579.

Itô, J., and A. Mester. 2012. “Recursive Prosodic Phrasing in Japanese.” In Prosody Matters: Essays in Honor of Elisabeth Selkirk, edited by T. Borowsky et al.

Katahira, K., K. Suzuki, H. Kagawa, and K. Okanoya. 2013. “A Simple Explanation for the Evolution of Complex Song Syntax in Bengalese Finches.” Biology Letters 9 (6): 20130842.

Katahira, K., K. Suzuki, H. Kagawa, and K. Okanoya. 2013. “Data from: A Simple Explanation for the Evolution of Complex Song Syntax in Bengalese Finches.” Dataset. Zenodo.

Koch, H. 1983. Introductory Essay on Composition. Translated by N. K. Baker. New Haven: Yale University Press.

Krieg, C. A. 2024. “Data from: Sex Differences in the Neural Song Circuit and Its Relationship to Song Acoustic Complexity in House Wrens (Troglodytes aedon).” Dataset. Dryad.

Krieg, C. A., and J. Wade. 2023. “Sex Differences in the Neural Song Circuit and Its Relationship to Song Acoustic Complexity in House Wrens (Troglodytes aedon).” Brain, Behavior and Evolution 98 (5): 231–244.

Lachlan, R. F., and S. Nowicki. 2015. “Context-Dependent Categorical Perception in a Songbird.” Proceedings of the National Academy of Sciences 112 (6): 1892–1897.

Lameira, A. R., M. E. Hardus, A. Ravignani, T. Raimondi, and M. Gamba. 2023. “Recursive Self-Embedded Vocal Motifs in Wild Orangutans.” bioRxiv.

Lee, K., A. T. Zai, M. Basha, and R. Hahnloser. 2026b. “Zebra Finch Vocalization Dataset for Semi-Automated Clustering.” Dataset. Zenodo.

Lee, K., M. Basha, A. T. Zai, and R. H. R. Hahnloser. 2026a. “Benchmarking Automated and Semi-Automated Vocal Clustering Methods.” Applied Sciences 16 (2): 810.

Morita, T., and H. Koda. 2019. “Data from: Superregular Grammars Do Not Provide Additional Explanatory Power but Allow for a Compact Analysis of Animal Song.” Dataset. Dryad.

Morita, T., and H. Koda. 2019. “Superregular Grammars Do Not Provide Additional Explanatory Power but Allow for a Compact Analysis of Animal Song.” Royal Society Open Science 6 (7): 190139.

Prat, Y., M. Taub, E. Pratt, and Y. Yovel. 2017. “An Annotated Dataset of Egyptian Fruit Bat Vocalizations Across Varying Contexts and During Vocal Ontogeny.” Dataset. Figshare.

Prat, Y., M. Taub, E. Pratt, and Y. Yovel. 2017. “An Annotated Dataset of Egyptian Fruit Bat Vocalizations Across Varying Contexts and During Vocal Ontogeny.” Scientific Data 4: 170143.

Rey, A., P. Perruchet, and J. Fagot. 2012. “Centre-Embedded Structures Are a By-Product of Associative Learning and Working Memory Constraints: Evidence from Baboons (Papio papio).” Cognition 123 (1): 180–184.

Rothstein, W. 1989. Phrase Rhythm in Tonal Music. New York: Schirmer Books.

Schall, E., K. Thomisch, O. Boebel, G. Gerlach, S. Mangia Woods, I. Roca, and I. Van Opzeeland. 2021b. “Humpback Whale Song Presence in the Atlantic Sector of the Southern Ocean.” Dataset. Dryad.

Schall, E., K. Thomisch, O. Boebel, G. Gerlach, S. Mangia Woods, I. T. Roca, and I. Van Opzeeland. 2021a. “Humpback Whale Song Recordings Suggest Common Feeding Ground Occupation by Multiple Populations.” Scientific Reports 11: 18806.

Schreuder, M. n.d. “Recursion in Phonology.” Dissertation chapter, extended version of Schreuder and Gilbers 2004.

Schreuder, M., D. Gilbers, and H. Quéné. 2009. “Recursion in Phonology.” Lingua.

Selkirk, E. 1995. “The Prosodic Structure of Function Words.” In Papers in Optimality Theory, edited by J. Beckman et al.

Selkirk, E. 2011. “The Syntax–Phonology Interface.” In The Handbook of Phonological Theory, 2nd ed., edited by J. Goldsmith, J. Riggle, and A. Yu. Oxford: Blackwell.

Stafford, K. M., M. Castellote, M. Guerra, and C. L. Berchok. 2018. “Data from: Extreme Diversity in the Songs of Spitsbergen’s Bowhead Whales.” Dataset. Dryad.

Stafford, K. M., M. Castellote, M. Guerra, and C. L. Berchok. 2018. “Extreme Diversity in the Songs of Spitsbergen’s Bowhead Whales.” Biology Letters 14 (4): 20180056.

Suzuki, T. N. 2024. “Animal Linguistics.” Annual Review of Ecology, Evolution, and Systematics 55: 205–226.

Vernes, S. C., et al. 2021. “The Multi-Dimensional Nature of Vocal Learning.” Philosophical Transactions of the Royal Society B 376 (1836).

Vigário, M. 2010. “Prosodic Structure Between the Prosodic Word and the Phonological Phrase.”

Vogel, I. 2009, 2012, 2019. “Recursion in Prosodic Structure.”

Wang, T. S., X. Li, C. Zhang, M. Wu, and K. Q. Zhu. 2024. “Phonetic and Lexical Discovery of Canine Vocalization.” Findings of the Association for Computational Linguistics: EMNLP 2024.

Xeno-canto Foundation. n.d. “Xeno-canto: Sharing Bird and Wildlife Sounds from Around the World.” Accessed 2026. https://www.xeno-canto.org.

Yip, M. J. 2006. “The Search for Phonology in Other Species.” Trends in Cognitive Sciences 10 (10): 442–446.